Term Project — Dataset Selection Report

Dataset Chosen: Olympics Data

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Dataset Overview

Our group has selected the **Olympics** dataset. It contains **271,116 athlete-event records** spanning **1896 to 2016**, covering **135,571 unique athletes**, **66 sports**, **765 distinct events**, and **230 national Olympic committees (NOCs)** across both Summer (222,552 records) and Winter (48,564 records) games.

Each row represents one athlete competing in one event at one games. The fifteen variables combine athlete demographics (`sex`, `age`, `height`, `weight`), competition identifiers (`games`, `year`, `season`, `city`), classification (`sport`, `event`, `team`, `noc`), and the outcome variable `medal`.

We chose this dataset because it supports both descriptive and inferential work: it is large enough for meaningful subgroup analysis, has a long time dimension, and contains a natural categorical response variable alongside continuous predictors.

Proposed Research Questions

1. Do physical attributes predict medal success, and does this vary by sport?

Mean male height ranges from 167.6 cm in gymnastics to 194.9 cm in basketball. We would test whether height, weight, and age predict medaling within a given sport, and whether the direction of that relationship reverses across sports — supporting a logistic regression with medal status as a binary outcome.

2. How has female participation changed over time, and has the gap closed?

Female representation in the Summer Games rose from 1.7% in 1900 to 45.5% in 2016. We would model this trajectory, identify which sports drove the change, and test whether parity has been reached uniformly or remains concentrated in particular sports and countries.

3. What explains differences in national medal performance?

The United States leads with 5,637 total medals, followed by the former Soviet Union (2,503) and Germany (2,165). We would examine whether medal counts are better explained by delegation size, sport specialization, or host-country advantage.

4. Is there a measurable host-country advantage?

Using within-country comparisons across games, we would test whether a nation's medal rate increases in years it hosts relative to its own baseline, controlling for delegation size.

Anticipated Data Challenges

- **Missingness in physical attributes.** Height is missing in 22.2% of records and weight in 23.2%, concentrated in earlier games. Any analysis using these variables will require either restriction to a modern subset or an explicit missing-data strategy.
- **Medal variable encoding.** 85.3% of records have `NA` for `medal`. These are non-medaling entries rather than missing data, so the variable must be recoded to a binary indicator before modeling.
- **Team-event double counting.** Every member of a winning team receives a medal row, so raw medal counts overweight team sports relative to individual events. Country-level analysis will require deduplicating by event.
- **Changing NOC definitions.** Historical entities such as the URS, GDR, and FRG complicate longitudinal country comparisons and will need explicit handling.